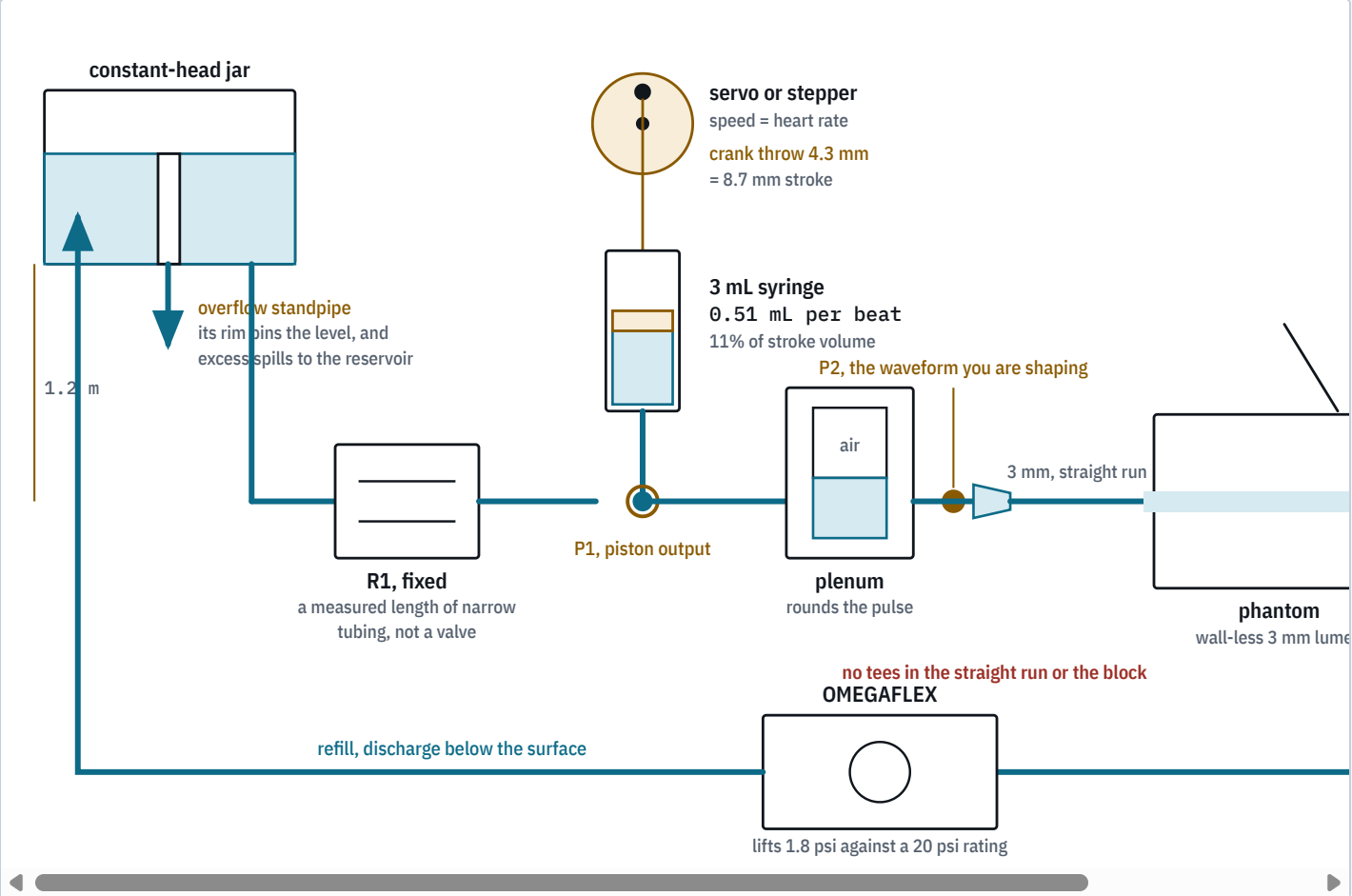
GW BME CAPSTONE · TCD PHANTOM · GROUP 2

Side-Branch Pulse Generator

With a constant-head jar supplying steady flow, the pulse gets added downstream by a piston that shuttles about half a millilitre per beat. It never carries the flow, only the variation, which is why it is so small.

1 Where it sits

The pump lifts fluid to the jar and does nothing else. Everything that shapes the waveform is downstream of it.

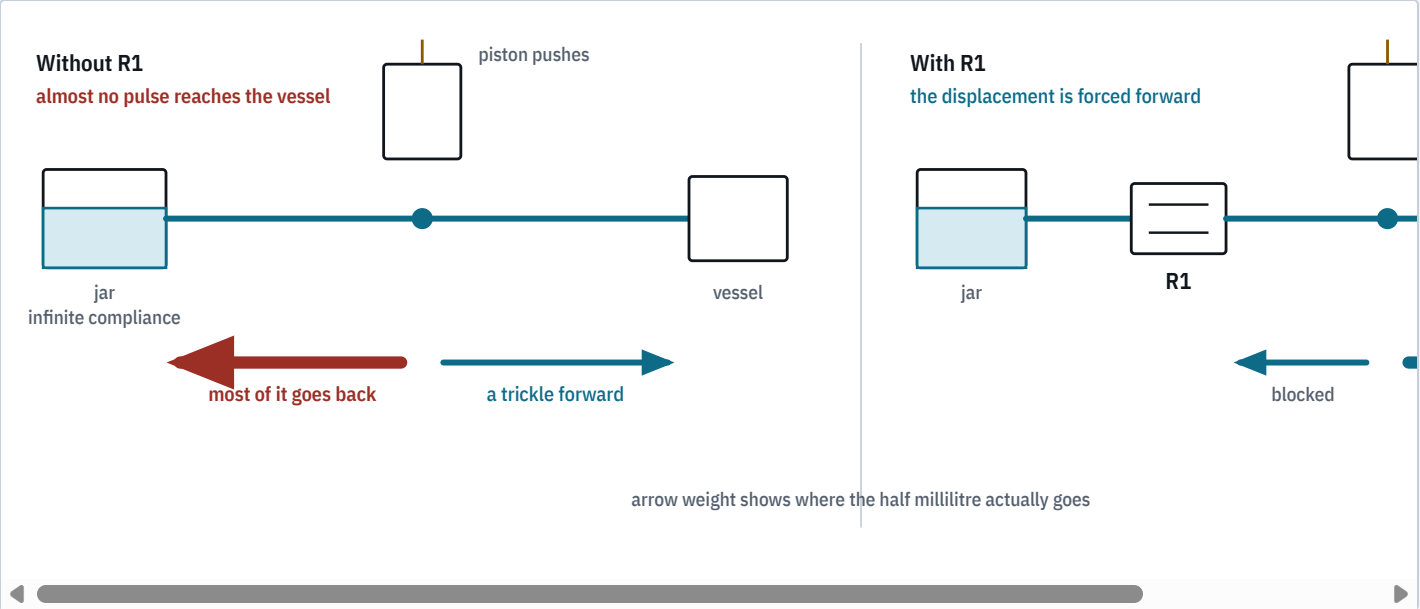


The pump never touches the waveform. It lifts fluid to the jar, the weir pins the level, and the free surface blocks everything upstream, including the pump's own roller pulses. That is what frees you to use the OMEGAFLEX and to set heart rate independently of flow.

Drawing scrolls sideways.

2 Why R1 has to be there

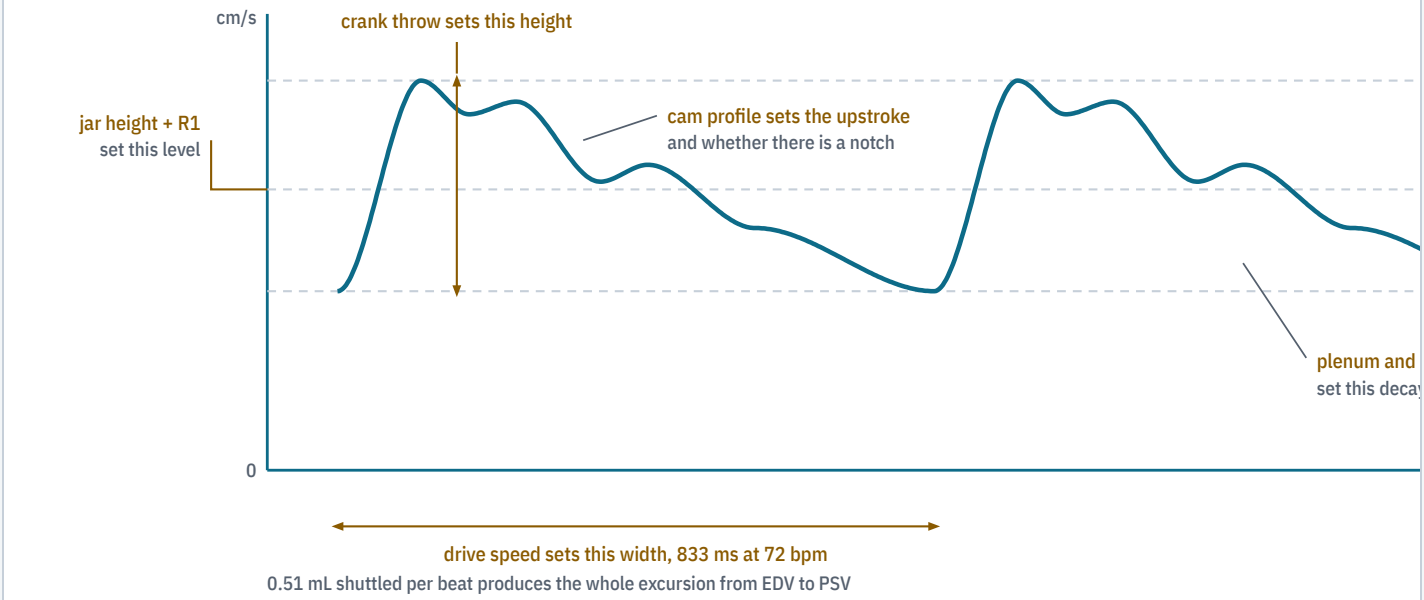
The single detail most likely to be left out, and it makes the difference between a pulse and nothing.



R1 much larger than R2 is the condition. The jar is a free surface, so it will absorb anything offered to it without its pressure moving. R1 is what makes the path backward expensive and the path forward cheap. Leave it out and the rig runs, quietly, with almost no pulsation, and the cause is not obvious from looking at it.

3 Which knob does what

Five controls, five features, and none of them fight each other. That is what no pump on the inventory could offer.



Heart rate and cardiac output are finally separate knobs. On any peristaltic they are the same knob, because beats per minute is rollers times RPM and flow is displacement times RPM. Here the drive motor sets rate, the jar and R1 set flow, and neither one touches the other.

CONTROL	SETS	VALUE FOR A NORMAL MCA
Jar height and R1	Mean flow	`324 mL/min, MFV 76 cm/s`
Crank throw	Pulse amplitude	`4.3 mm throw, 0.51 mL/beat`
Drive speed	Heart rate	`72 bpm`
Cam profile	Upstroke and notch	`117 ms ejection`
Plenum air volume and R2	Diastolic decay, so PI	`PI 0.85`

It is small because it only makes the variation.

Stroke volume through the vessel is 4.5 mL per beat, but the steady source already supplies the mean. The piston only has to redistribute flow in time, which is 0.51 mL, about 11%. A 3 mL

syringe at 8.7 mm of stroke, under a newton of force, moving at 26 mm/s. That is a hobby servo on a crank.

This is not the stepper piston pump that failed in the literature.

A Clemson thesis and a Groningen group both built stepper-driven piston pumps that had to carry the whole circulation, and both fell short of their requirements. This carries no net flow, needs no check valves, needs no priming and sees almost no load. Different job.

A cam is a waveform you can hold in your hand.

Because the shape comes from a printed profile rather than from pump physics, a second cam gives a second waveform. Normal, high pulsatility, blunted. That is a far more useful phantom than one that produces whatever its pump happens to produce, and it lets the rig replay a measured patient waveform.

Pressure is the tuning instrument, Doppler is the acceptance test.

P2 sits at the last place you can tee without disturbing the straight run, so it shows the pressure waveform arriving at the vessel. It responds instantly and is clean, which is what you want while cutting cams. But it is not the answer: the velocity waveform inside the lumen is, and the taper, the run and the vessel's own compliance all act between them. You cannot tee into a wall-less cast lumen, so the Doppler is the only instrument that sees inside the block. Tune against P2, accept against the Doppler. P1 exists so that when the pulse is too small you can tell whether the piston is underdelivering or the line is eating it.

What still has to be measured.

The relationship between cam profile and the velocity that comes out is not one to one, because the plenum, the vessel compliance and R2 all act on it. Expect to iterate the cam against the Doppler trace. That iteration is the project.

Pulser sized for a 3 mm lumen at EDV 45 and PSV 110 cm/s, 72 bpm, which gives MFV 76 cm/s and PI 0.85. Velocity bands are Group 2's own. Displacement and force are derived from those, not measured, and want checking on the bench.